

Software Requirement Specification (SRS)

Hospital Management System

1. Introduction

The Hospital Management System (HMS) is a software application used to manage the daily activities of a hospital. It helps to maintain patient details, doctor details, appointments, medical records, billing and payment information.

Purpose:

- Reduce manual paperwork.
- Store hospital information digitally.
- Manage patient and doctor information easily.
- Provide quick access to medical and appointment records.
- Improve the efficiency of hospital operations.

2. Functional Requirements

The system should provide the following functions:

2.1 Patient Management

- Patient registration
- Store patient personal details
- Generate patient ID
- View and update patient records

2.2 Doctor Management

- Add doctor details
- View doctor information
- Manage doctor availability and schedules

2.3 Appointment Management

- Book appointments
- Check doctor availability
- View and cancel appointments

2.4 Medical Record Management

- Store patient medical history
- Doctor can add diagnosis and treatment
- Doctor can prescribe medicines

2.5 Billing and Payment

- Generate patient bills
- Calculate treatment charges
- Record payment details
- Display payment status

2.6 Login and Logout

- User login with username and password
- Different access for Admin, Doctor, Receptionist and Patient
- Secure logout

3. Interface Requirements

3.1 GUI (Graphical User Interface)

The system should have a simple and user-friendly interface containing:

- Login page
- Registration page
- Patient dashboard
- Doctor dashboard
- Appointment page
- Medical record page
- Billing page
- Admin dashboard
- Logout option

3.2 Hardware Interface

The system requires:

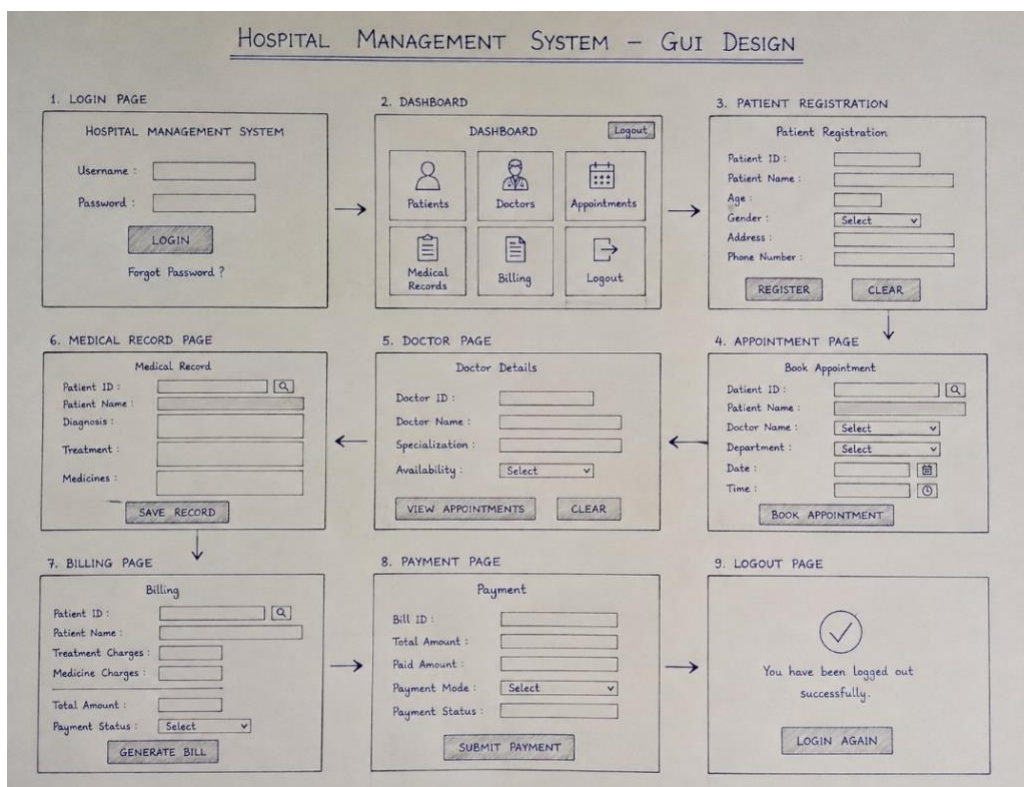
- Computer/Laptop
- Keyboard
- Mouse
- Monitor
- Printer for printing bills/reports

- Internet/LAN connection

3.3 Software Interface

The system can use:

- Operating System: Windows/Linux
- Front End: HTML, CSS, JavaScript
- Back End: Java/Python/PHP
- Database: MySQL
- Web Browser: Chrome/Firefox



4. Non-Functional Requirements

4.1 Security

- Only authorized users can access the system.
- Users must log in using a username and password.
- Patient information should be protected.
- Different users should have different access permissions.
- Passwords should be stored securely.

4.2 Reliability

- The system should work correctly without frequent failures.
- Patient and medical records should not be lost.
- Regular database backup should be maintained.
- The system should provide accurate billing and appointment information.

4.3 Availability

- The system should be available during hospital working hours.
- Users should be able to access the system whenever required.
- Backup and recovery facilities should be provided in case of system failure.

4.4 Performance

- The system should respond quickly to user requests.
- Patient records and appointments should load within a short time.
- The system should support multiple users.

4.5 Usability

- .The interface should be simple and easy to understand.
- Users should be able to navigate between pages easily.
- Error messages should be clear.

4.6 Maintainability

- The system should be easy to update.
- Database and software components should be easy to maintain.
- New features should be added without affecting existing functions.

5. Conclusion

The Hospital Management System provides an efficient way to manage patients, doctors, appointments, medical records, billing and payments. It reduces paperwork, improves data security and makes hospital operations easier and faster.